

EX: 1 Design a visiting card using layout components.

XML Layout (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

    xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:gravity="center"

    android:orientation="vertical"

    android:background="#E3F2FD"

    android:padding="20dp">

    <TextView

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="VISITING CARD"

        android:textSize="24sp"

        android:textStyle="bold"

        android:textColor="#000"/>

    <TextView

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Dhivya Sri
```

```
        android:textSize="22sp"

        android:textStyle="bold"

        android:layout_marginTop="20dp"/>
<TextView

    android:layout_width="wrap_content"

    android:layout_height="wrap_content"

    android:text="Software Developer"/>

<TextView

    android:layout_width="wrap_content"

    android:layout_height="wrap_content"

    android:text="Phone: +91 9876543210"/>

<TextView

    android:layout_width="wrap_content"

    android:layout_height="wrap_content"

    android:text="Email: dhivya@example.com"/>

<TextView

    android:layout_width="wrap_content"

    android:layout_height="wrap_content"

    android:text="Address: Chennai, Tamil Nadu"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.visitingcard;
```

```
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

    }

}
```

EX: 2 Create an android application to build a basic calculator using buttons, textview and edit text

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">
    <EditText
        android:id="@+id/num1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter First Number"
        android:inputType="numberDecimal"/>
    <EditText
        android:id="@+id/num2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Second Number"
        android:inputType="numberDecimal"/>
    <Button
        android:id="@+id/add"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
```

```
        android:text="Addition"/>
    <Button
        android:id="@+id/sub"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Subtraction"/>
    <Button
        android:id="@+id/mul"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Multiplication"/>
    <Button
        android:id="@+id/div"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Division"/>
    <TextView
        android:id="@+id/result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result"
        android:textSize="20sp"
        android:paddingTop="20dp"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.calculator;
```

```
import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

import android.widget.*;

import android.view.View;

public class MainActivity extends AppCompatActivity {

    EditText num1, num2;

    Button add, sub, mul, div;

    TextView result;

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        num1 = findViewById(R.id.num1);

        num2 = findViewById(R.id.num2);

        add = findViewById(R.id.add);

        sub = findViewById(R.id.sub);

        mul = findViewById(R.id.mul);

        div = findViewById(R.id.div);

        result = findViewById(R.id.result);

        add.setOnClickListener(v -> {

            double a = Double.parseDouble(num1.getText().toString());

            double b = Double.parseDouble(num2.getText().toString());

            result.setText("Result = " + (a + b));

        });

        sub.setOnClickListener(v -> {

            double a = Double.parseDouble(num1.getText().toString());

            double b = Double.parseDouble(num2.getText().toString());
```

```
        result.setText("Result = " + (a - b));
    });

    mul.setOnClickListener(v -> {
        double a = Double.parseDouble(num1.getText().toString());
        double b = Double.parseDouble(num2.getText().toString());
        result.setText("Result = " + (a * b));
    });

    div.setOnClickListener(v -> {
        double a = Double.parseDouble(num1.getText().toString());
        double b = Double.parseDouble(num2.getText().toString());
        result.setText("Result = " + (a / b));
    });
}
}
```

EX : 3 Sign-Up Activity with Username and Password Validation

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/etUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"/>

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword"/>

    <EditText
        android:id="@+id/etConfirmPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Confirm Password"
        android:inputType="textPassword"/>
```

```
<Button
    android:id="@+id/btnSignup"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Sign Up"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.signup

import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.Toast

class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
        val username = findViewById<EditText>(R.id.etUsername)
        val password = findViewById<EditText>(R.id.etPassword)
        val confirmPassword = findViewById<EditText>(R.id.etConfirmPassword)
        val signup = findViewById<Button>(R.id.btnSignup)
        signup.setOnClickListener {
            val user = username.text.toString()
            val pass = password.text.toString()
            val confirm = confirmPassword.text.toString()
            if (user.isEmpty() || pass.isEmpty() || confirm.isEmpty()) {
```

```
        Toast.makeText(this, "All fields are required",
Toast.LENGTH_SHORT).show()

    }

    else if (pass.length < 8) {

        Toast.makeText(this, "Password must contain at least 8 characters",
Toast.LENGTH_SHORT).show()

    }

    else if (pass != confirm) {

        Toast.makeText(this, "Passwords do not match",
Toast.LENGTH_SHORT).show()

    }

    else {

        Toast.makeText(this, "Sign-Up Successful",
Toast.LENGTH_LONG).show()

    }

    }

    }

}
```

EX :4 Android App to Change Wallpaper Randomly Every 30 Seconds

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
    <Button
        android:id="@+id/btnWallpaper"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Start Wallpaper Change"/>
</LinearLayout>
```

MainActivity.KT

```
package com.example.wallpaperapp
import android.app.WallpaperManager
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.os.Handler
import android.widget.Button
import java.util.*
import androidx.core.content.res.ResourcesCompat
class MainActivity : AppCompatActivity() {
    private val handler = Handler()
```

```

private val wallpapers = arrayOf(
    R.drawable.wallpaper1,
    R.drawable.wallpaper2,
    R.drawable.wallpaper3
)

private val runnable = object : Runnable {
    override fun run() {
        val random = Random()
        val image = wallpapers[random.nextInt(wallpapers.size)]
        val wallpaperManager =
WallpaperManager.getInstance(applicationContext)
        val bitmap = ResourcesCompat.getDrawable(resources, image, null)
        ?.let {
            (it as android.graphics.drawable.BitmapDrawable).bitmap
        }
        if (bitmap != null) {
            wallpaperManager.setBitmap(bitmap)
        }
        handler.postDelayed(this, 30000)
    }
}

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_main)
    val button = findViewById<Button>(R.id.btnWallpaper)
    button.setOnClickListener {

```

```
        handler.post(runnable)
    }
}
override fun onDestroy() {
    super.onDestroy()
    handler.removeCallbacks(runnable)
}
}
```

AndroidManifest.xml

```
<uses-permission android:name="android.permission.SET_WALLPAPER"/>
<application
    ...
</application>
```